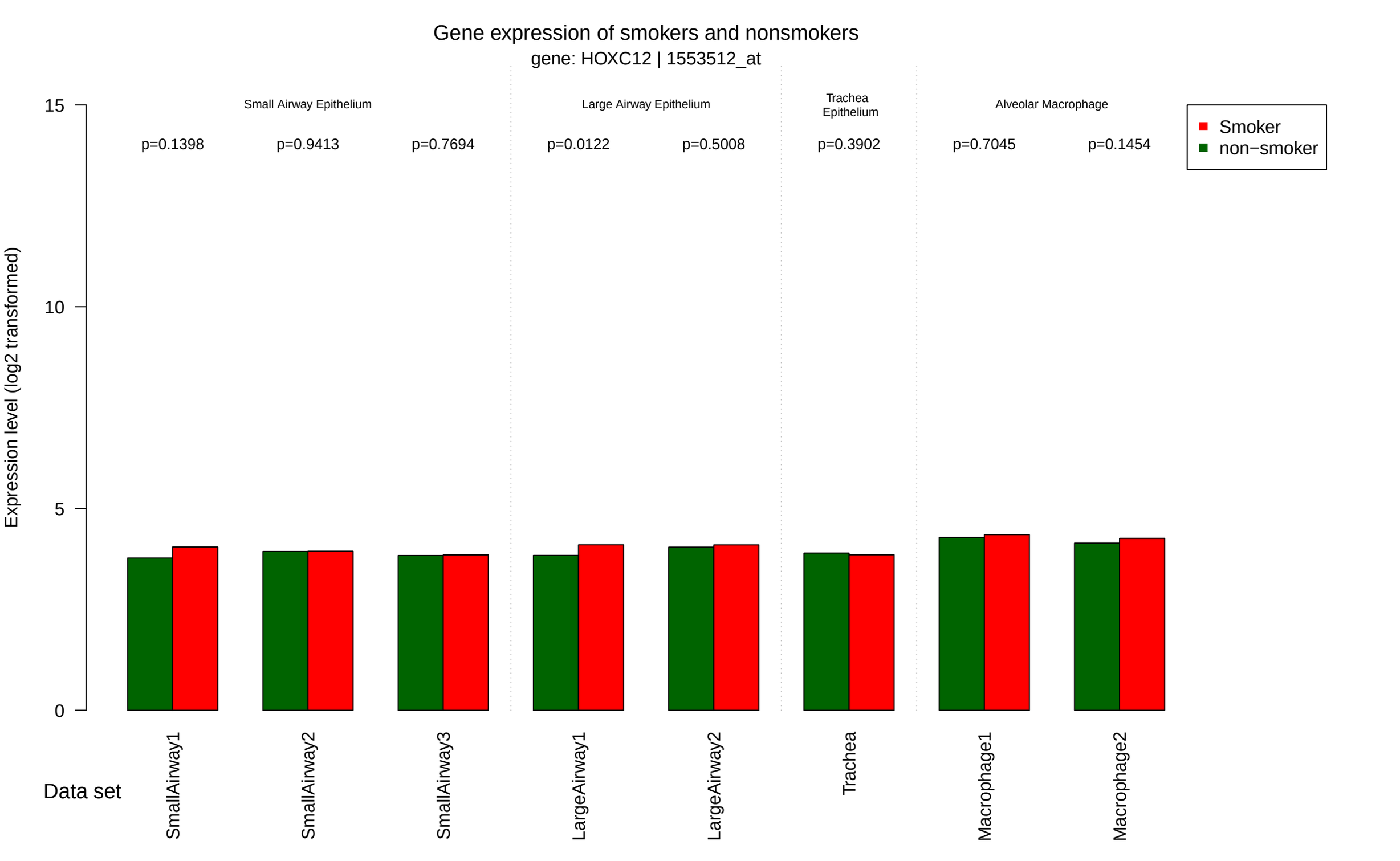
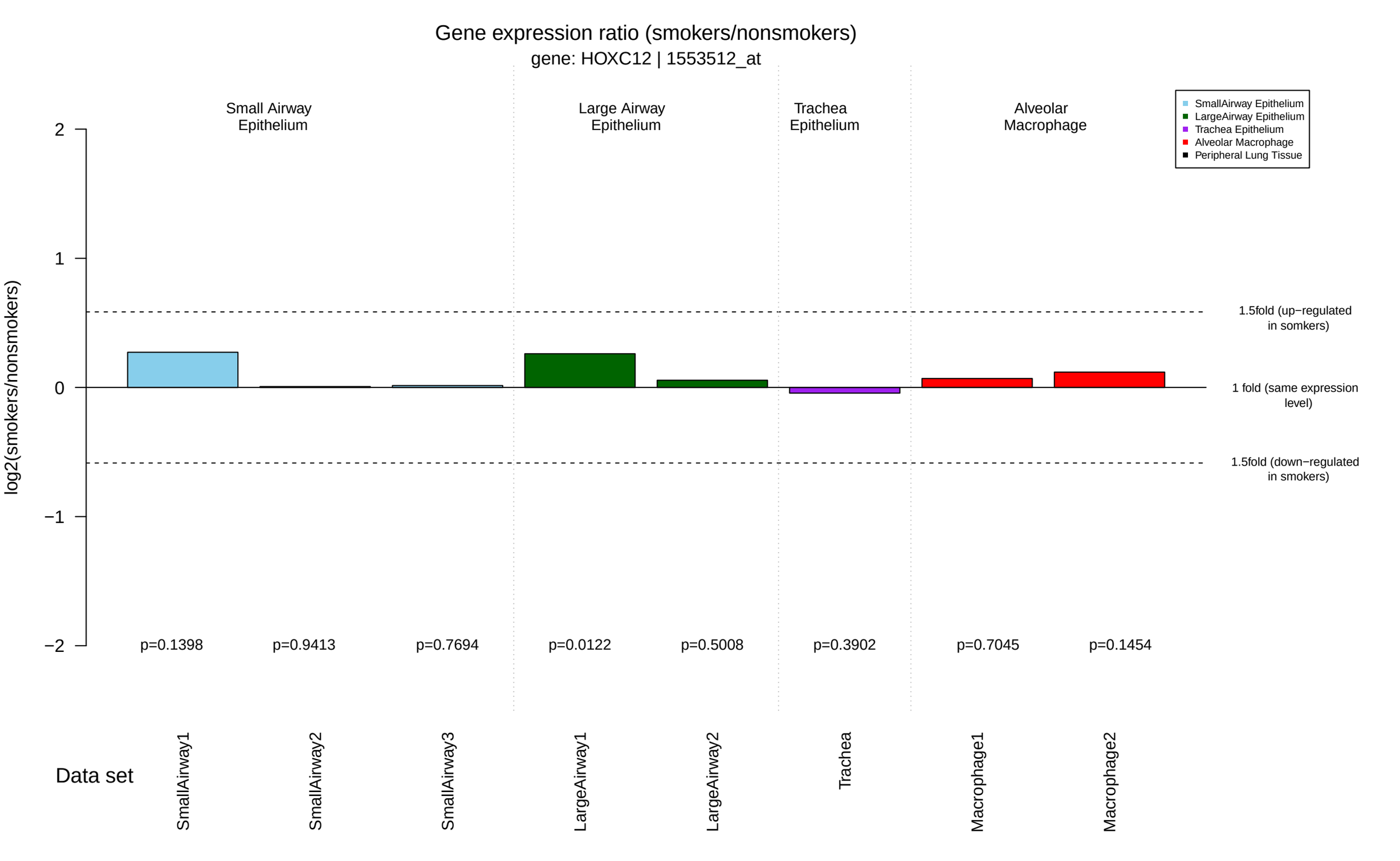
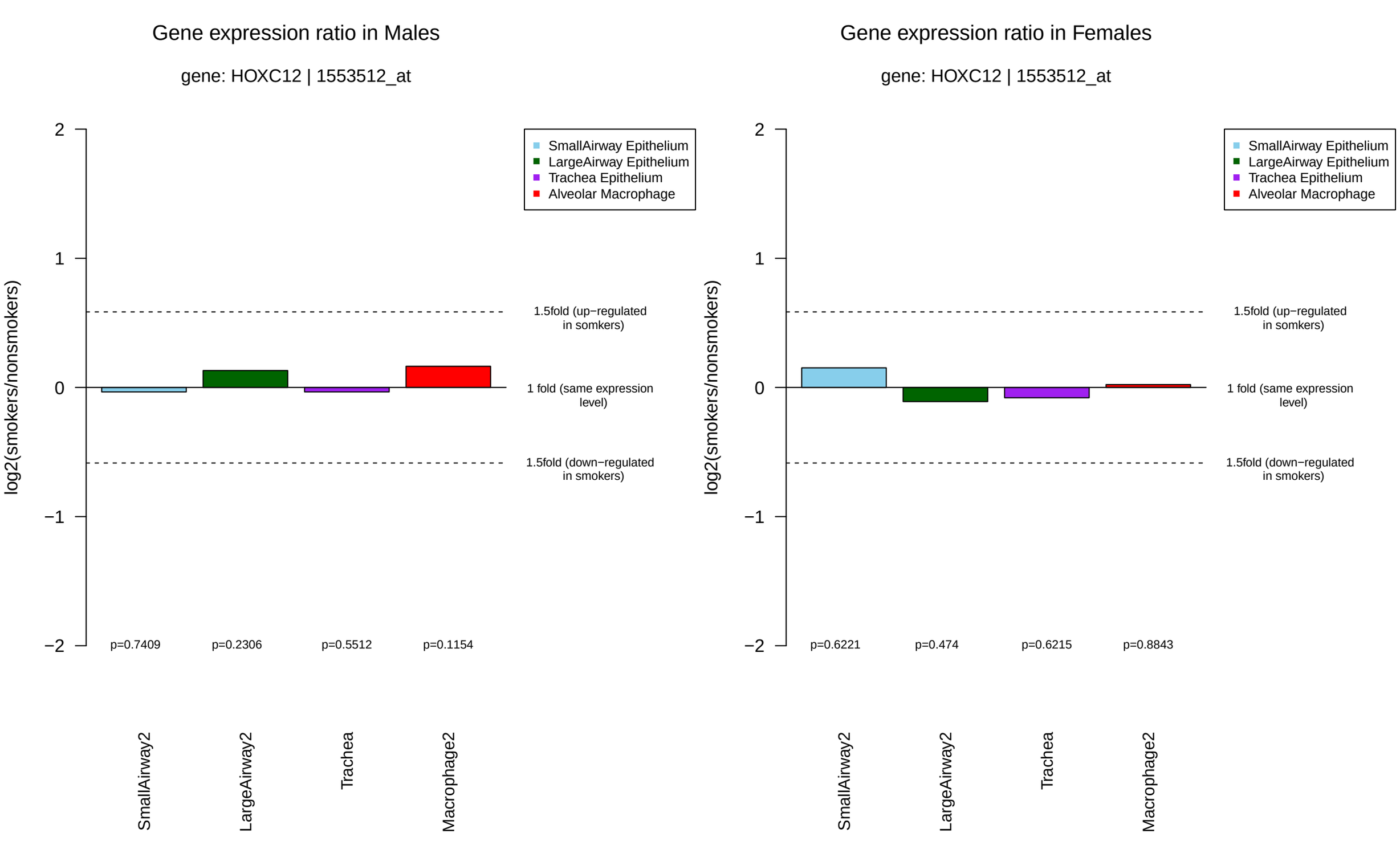
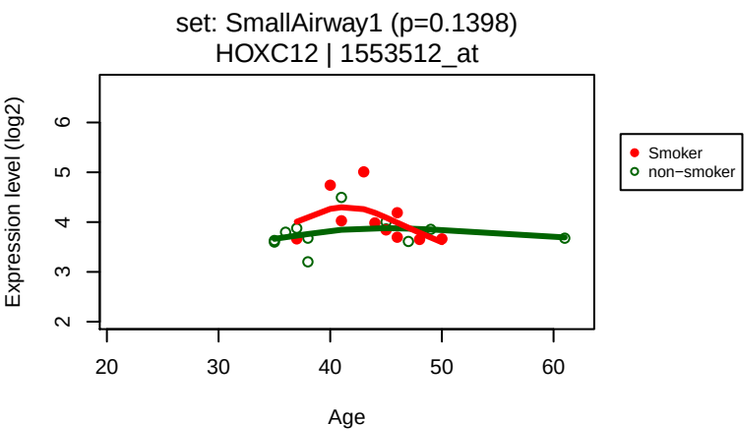
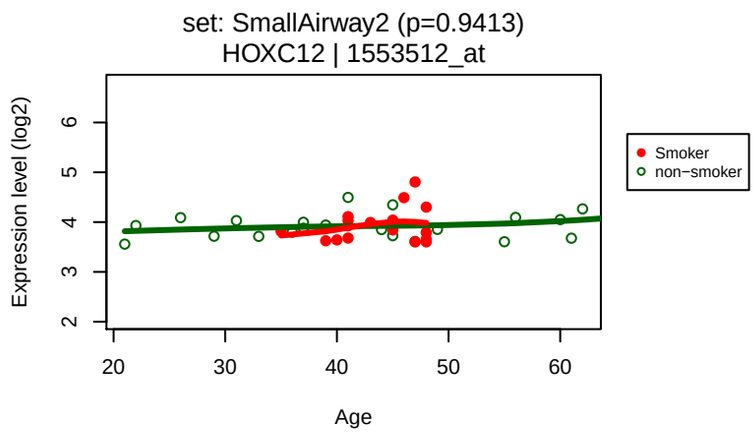
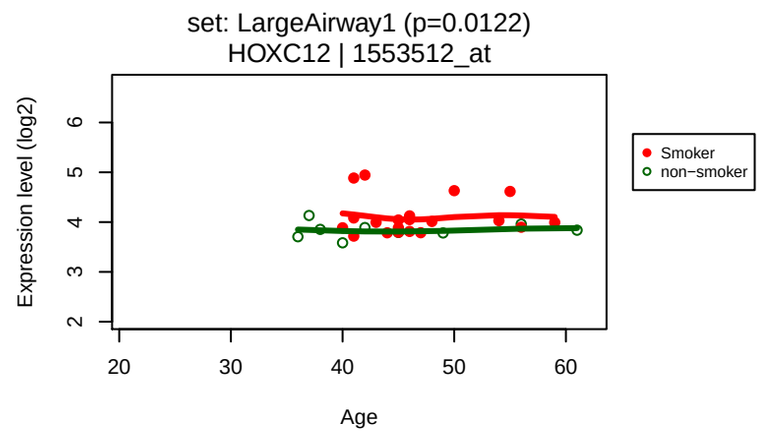

Gene expression of smokers and nonsmokers
gene: HOXC12 | 1553512_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1398
p=0.9413
p=0.7694
p=0.0122
p=0.5008
p=0.3902
p=0.7045
p=0.1454
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

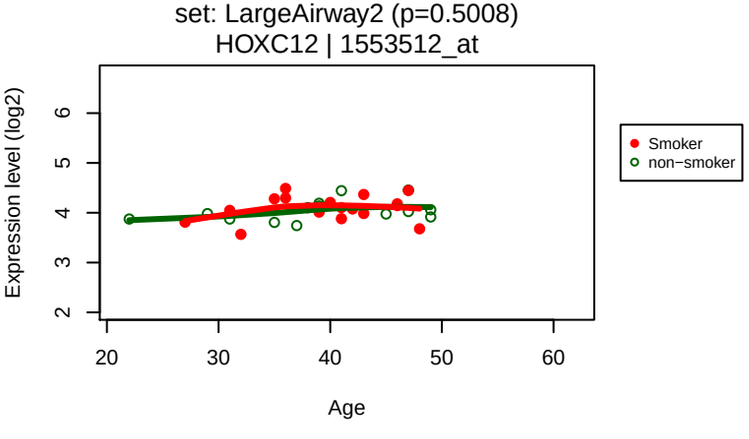

Gene expression ratio (smokers/nonsmokers)
gene: HOXC12 | 1553512_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1398
p=0.9413
p=0.7694
p=0.0122
p=0.5008
p=0.3902
p=0.7045
p=0.1454
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

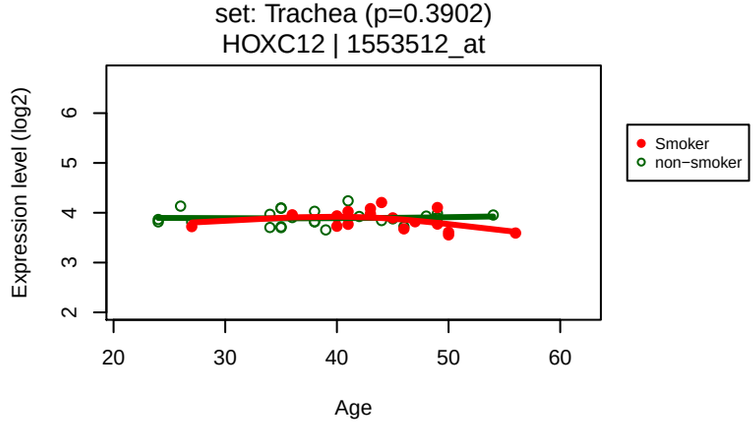

Gene expression ratio in Males
gene: HOXC12 | 1553512_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7409
p=0.2306
p=0.5512
p=0.1154
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXC12 | 1553512_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6221
p=0.474
p=0.6215
p=0.8843
SmallAirway2
LargeAirway2
Trachea
Macrophage2

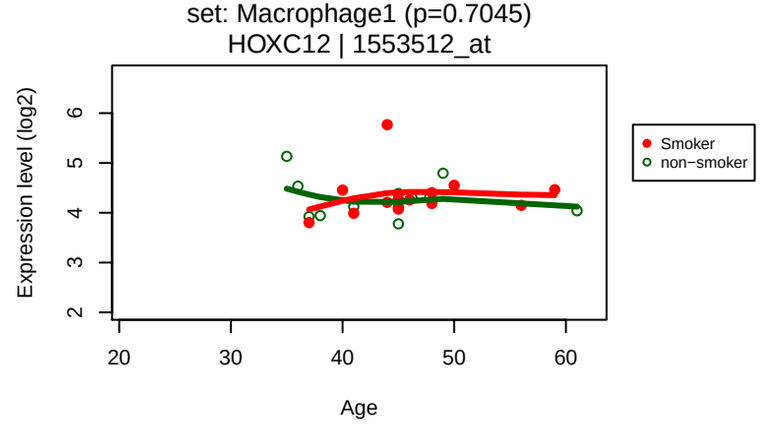

set: SmallAirway1 (p=0.1398)
HOXC12 | 1553512_at
Expression level (log2)
Age
Smoker
non−smoker

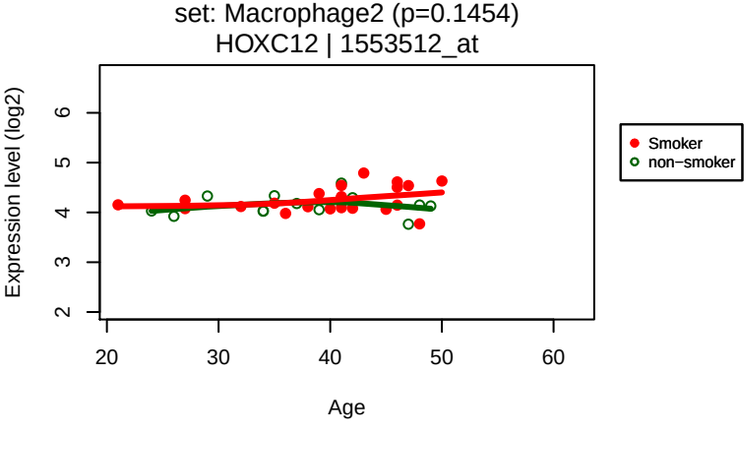

set: SmallAirway2 (p=0.9413)
HOXC12 | 1553512_at
Expression level (log2)
Age
Smoker
non−smoker

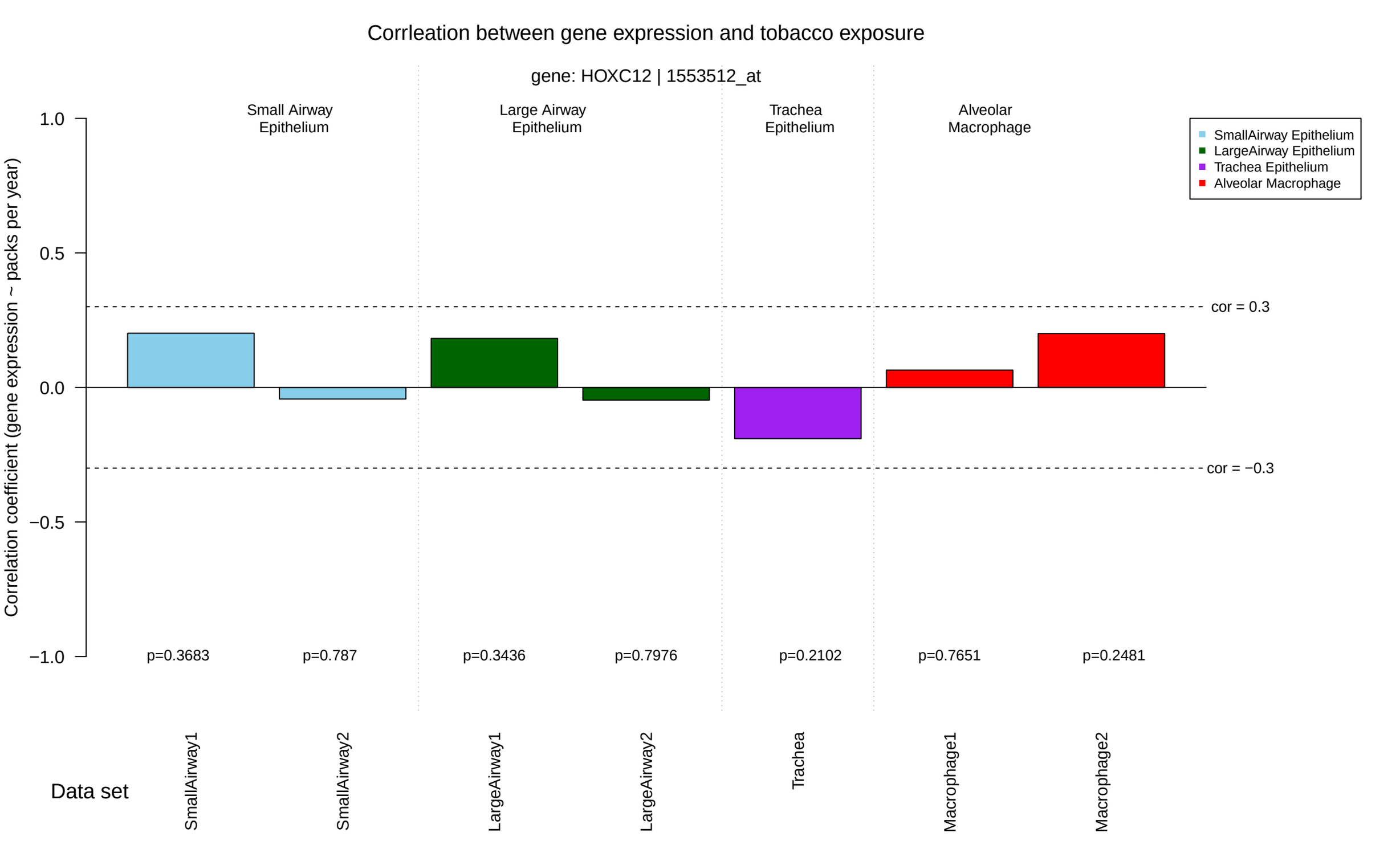

set: LargeAirway1 (p=0.0122)
HOXC12 | 1553512_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5008)
HOXC12 | 1553512_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3902)
HOXC12 | 1553512_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7045)
HOXC12 | 1553512_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1454)
HOXC12 | 1553512_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXC12 | 1553512_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3683
p=0.787
p=0.3436
p=0.7976
p=0.2102
p=0.7651
p=0.2481
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2